

BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `c:\Users\Donny\OneDrive\Post-Grad\Courses\BEE 5750\Homework 2\hw2-d
Installed JpegTurbo_jll          v3.1.3+0
Installed InvertedIndices        v1.3.1
Installed SentinelArrays        v1.4.8
Installed WorkerUtilities        v1.6.1
Installed InlineStrings          v1.4.5
Installed Roots                  v2.2.10
Installed Accessors              v0.1.42
Installed DataFrames             v1.8.0
Installed WeakRefStrings         v1.4.2
Installed TableTraits            v1.0.1
Installed ConstructionBase       v1.6.0
Installed DataValueInterfaces    v1.0.0
Installed PooledArrays           v1.4.3
Installed Ghostscript_jll        v9.55.1+0
Installed CompositionsBase       v0.1.2
Installed PrettyTables           v3.0.8
Installed CommonSolve            v0.2.4
Installed InverseFunctions        v0.1.17
Installed Crayons                 v4.1.1
Installed IteratorInterfaceExtensions v1.0.0
Installed Tables                 v1.12.1
Installed Latexify                v0.16.10
Installed Glib_jll                v2.86.0+0
Installed FilePathsBase          v0.9.24
Installed StringManipulation      v0.4.1
Installed CSV                     v0.10.15
Precompiling project...
16300.7 ms  IteratorInterfaceExtensions
 8681.3 ms  WorkerUtilities
 8671.5 ms  InverseFunctions
 8662.6 ms  CommonSolve
 8637.1 ms  InlineStrings
 8654.0 ms  DataValueInterfaces
 8630.7 ms  CompositionsBase
 8617.9 ms  InvertedIndices
 8601.8 ms  ConstructionBase
 8559.3 ms  PooledArrays
 8573.2 ms  Crayons
 8545.4 ms  SentinelArrays
 802.2 ms   TableTraits
 825.2 ms   LogExpFunctions → LogExpFunctionsInverseFunctionsExt
 846.6 ms   InverseFunctions → InverseFunctionsDatesExt

```

831.8 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
892.7 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
960.1 ms	InlineStrings → ParsersExt
1347.9 ms	JpegTurbo_jll
1546.4 ms	InverseFunctions → InverseFunctionsTestExt
1617.5 ms	FilePathsBase
1357.3 ms	Unitful → InverseFunctionsUnitfulExt
1252.5 ms	Unitful → ConstructionBaseUnitfulExt
2250.1 ms	Glib_jll
1133.9 ms	FilePathsBase → FilePathsBaseMmapExt
1498.2 ms	WeakRefStrings
1523.9 ms	Tables
2928.4 ms	StringManipulation
1486.4 ms	Libtiff_jll
1855.5 ms	Ghostscript_jll
1781.8 ms	FilePathsBase → FilePathsBaseTestExt
3198.6 ms	Accessors
2053.2 ms	Cairo_jll
1377.0 ms	Accessors → LinearAlgebraExt
1587.0 ms	Accessors → TestExt
3724.8 ms	Qt6Base_jll
1631.6 ms	Accessors → UnitfulExt
2017.2 ms	HarfBuzz_jll
4576.3 ms	Latexify
1792.6 ms	libass_jll
1705.3 ms	Pango_jll
4473.0 ms	Roots
2275.1 ms	UnitfulLatexify
2458.3 ms	Latexify → SparseArraysExt
1365.2 ms	Roots → RootsUnitfulExt
5613.9 ms	Qt6ShaderTools_jll
3415.4 ms	FFMPEG_jll
2508.3 ms	FFMPEG
2996.9 ms	GR_jll
3643.2 ms	Qt6Declarative_jll
17747.0 ms	CSV
5173.9 ms	GR
30009.1 ms	PrettyTables
45487.7 ms	DataFrames
3066.6 ms	Latexify → DataFramesExt
73841.4 ms	Plots
10402.6 ms	Plots → UnitfulExt

57 dependencies successfully precompiled in 152 seconds. 162 already precompiled.

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg}/(1000\text{m}^3)$. You can do this analytically or computationally.

```
function concentration_segment1(x)
    return 5.172 - 3.500 * exp(-0.036 * x)
end

function concentration_segment2(x)
    return 4.286 - 0.490 * exp(-0.036 * (x - 15))
end

function get_concentration(x)
    if 0 <= x <= 15
        return concentration_segment1(x)
    elseif x > 15
        return concentration_segment2(x)
    else
        return NaN
    end
end

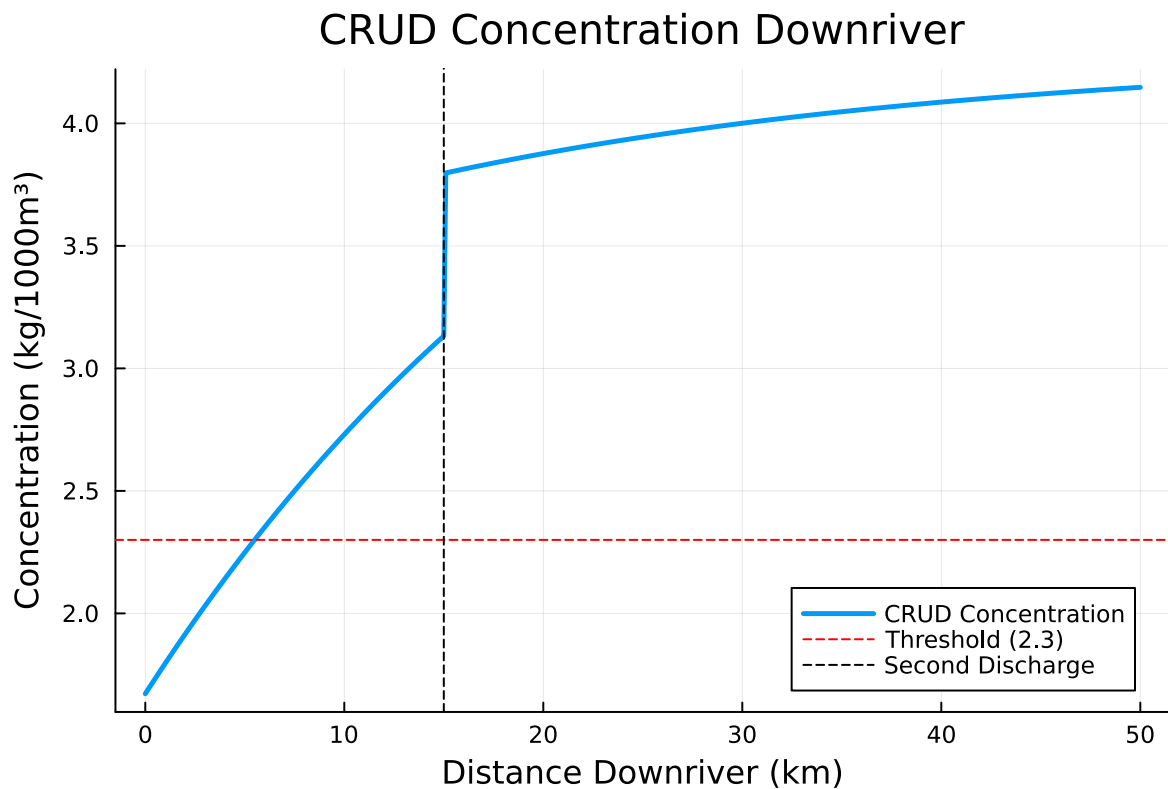
function plot_concentration_profile()
    # Define the distance ranges, concentrations, and threshold
    x_values = 0:0.1:50 # from 0 to 50 km, with a step of 0.1 km
    y_values = get_concentration.(x_values)
    threshold = 2.3

    # Plotting the concentration profile
    p = plot(x_values, y_values,
        label="CRUD Concentration",
        linewidth=2.5,
        title="CRUD Concentration Downriver",
        xlabel="Distance Downriver (km)",
        ylabel="Concentration (kg/1000m³)",
        legend=:bottomright
    )
    hline!(p, [threshold],
        linestyle=:dash,
        color=:red,
        label="Threshold (2.3)"
    )
end
```

```

vline!(p, [15],
        linestyle=:dash,
        color=:black,
        label="Second Discharge"
    )
display(p)
end
plot_concentration_profile()

```



Based on the plotting above, it can be inferred that after around 5 kilos from the starting point ($x=0$), the concentration of CRUD will always surpass the threshold ($2.3 \text{ kg}/1000\text{m}^3$). This shows that the decay rate of the CRUD cannot offset the amount and/or concentration of the discharges.

Thus, either the volume or the concentration of the CRUD need to be adjusted.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/\text{°C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
# Physical constants
const C = 51.0          # Heat capacity (J/m²/°C) - Scaled for a time step in years
const S = 1272.0        # Solar constant for Neoproterozoic (W/m²)
const A = 221.2         # Outgoing radiation coefficient (W/m²)
const B = -1.3          # Outgoing radiation coefficient (W/m²/°C)
const a_i = 0.5
const a_0 = 0.3

# Simulation parameters
const dt_years = 1.0    # Time step in years
const simulation_years = 200 # Total simulation time
const num_steps = Int(simulation_years / dt_years)
const initial_temps = -60.0:5.0:30.0 # Range of initial T (°C)

# Albedo function
function albedo(T::Float64)
    if T <= -10.0
        return a_i
    elseif -10.0 < T <= 10.0
        return a_i + (a_0 - a_i) * ((T + 10.0) / 20.0)
    else # T > 10.0
        return a_0
    end
end

# Main simulation
p = plot(
    title="Temperature Trajectories with Ice-Albedo Feedback",
    xlabel="Time (years)",
    ylabel="Global Mean Temperature (C)",
```



```

    legend=false,
    xlims=(0, simulation_years),
    ylims=(-70, 40)
)

for T0 in initial_temps
    temperatures = zeros(Float64, num_steps + 1)
    temperatures[1] = T0

    for i in 1:num_steps
        T_current = temperatures[i]

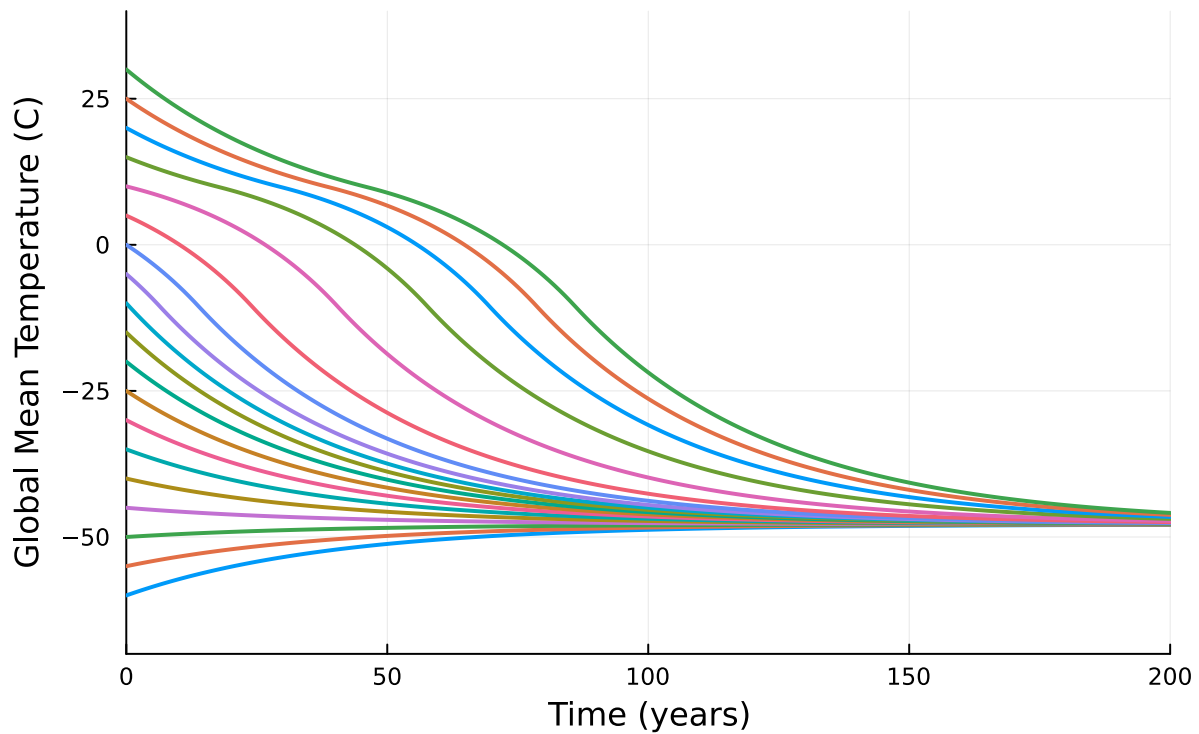
        incoming_rad = (1 - albedo(T_current)) * S / 4.0
        outgoing_rad = A - B * T_current
        net_radiation = incoming_rad - outgoing_rad
        T_next = T_current + (dt_years / C) * net_radiation
        temperatures[i+1] = T_next
    end

    time_axis = 0:dt_years:simulation_years
    plot!(p, time_axis, temperatures, lw=2)
end

println("Simulation complete. Displaying plot.")
display(p)

```

Temperature Trajectories with Ice-Albedo Feedback



Simulation complete. Displaying plot.

WARNING: redefinition of constant Main.S. This may fail, cause incorrect answers, or produce

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

```
# Physical constants
const C = 51.0          # Heat capacity (J/m2/°C) - Scaled for a time step in years
const S = 1368.0        # Solar constant for modern era (W/m2)
const A = 221.2         # Outgoing radiation coefficient (W/m2)
const B = -1.3          # Outgoing radiation coefficient (W/m2/°C)
const a_0 = 0.3

# Simulation parameters
```

```

const dt_years = 1.0                # Time step in years
const simulation_years = 200         # Total simulation time
const num_steps = Int(simulation_years / dt_years)
const initial_temps = -60.0:5.0:30.0 # Range of initial T (°C)

# Albedo function
function albedo(T::Float64)
    return a_0
end

# Main simulation
p = plot(
    title="Temperature Trajectories with Ice-Albedo Feedback",
    xlabel="Time (years)",
    ylabel="Global Mean Temperature (C)",
    legend=false,
    xlims=(0, simulation_years),
    ylims=(-70, 40)
)

for T0 in initial_temps
    temperatures = zeros(Float64, num_steps + 1)
    temperatures[1] = T0

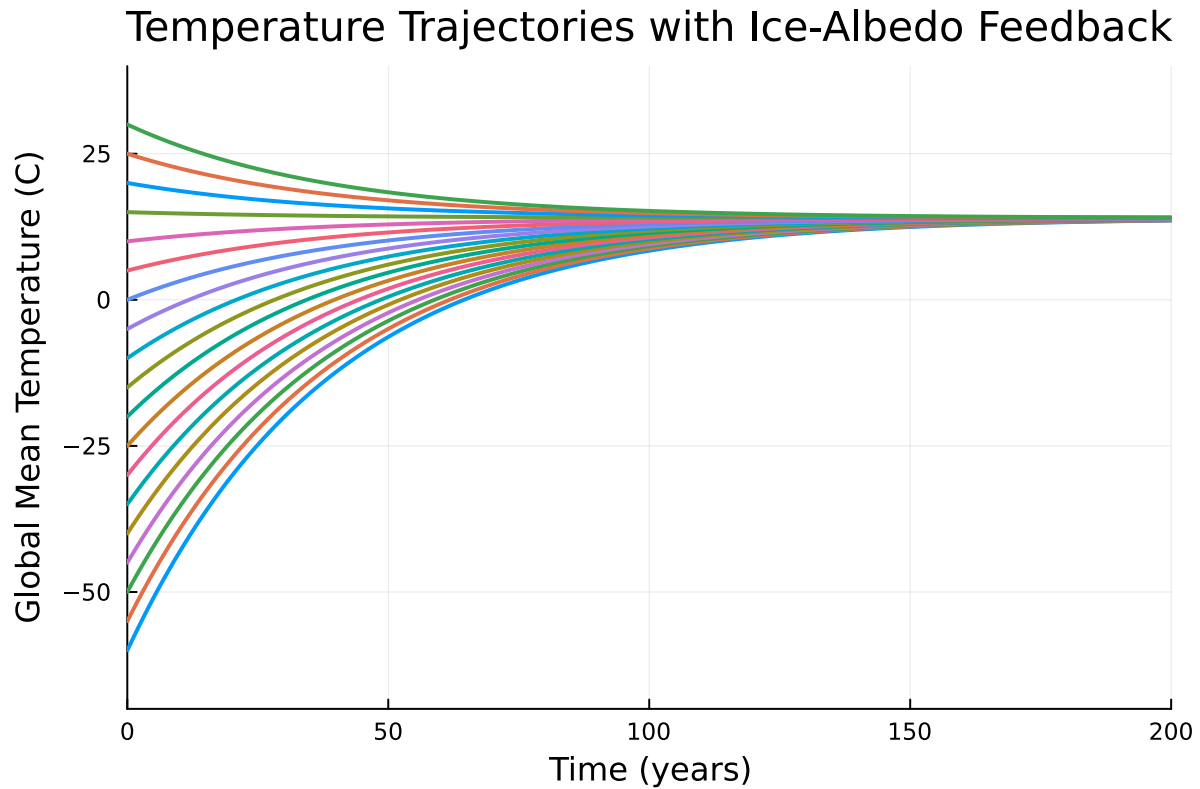
    for i in 1:num_steps
        T_current = temperatures[i]

        incoming_rad = (1 - albedo(T_current)) * S / 4.0
        outgoing_rad = A - B * T_current
        net_radiation = incoming_rad - outgoing_rad
        T_next = T_current + (dt_years / C) * net_radiation
        temperatures[i+1] = T_next
    end

    time_axis = 0:dt_years:simulation_years
    plot!(p, time_axis, temperatures, lw=2)
end

println("Simulation complete. Displaying plot.")
display(p)

```



Simulation complete. Displaying plot.

WARNING: redefinition of constant Main.S. This may fail, cause incorrect answers, or produce

According to the model being used, an increase in the solar radiation is capable of moving the earth from the Snowball Earth state. The graph above illustrates when the solar radiation (S) is 1368 W/m^2 and the $\alpha(a_0) = 0.3$ (less reflective surface). Across all ranges of initial temperature assumed ($-60^\circ\text{C} < T < 30^\circ\text{C}$), all approach the equilibrium of approximately 14°C .

References

LLM was used in this homework, specifically for problem 2.3 and 3.2 1. For problem 2.3, LLM was used to generate code for the plotting into the graph section. After the code is generated, user double check and debug the code if necessary 2. For problem 3.2, LLM was used to assist in building the loop for function and incorporating initial temperature limits. The code is then studied to learn the loop for structure and being debugged if necessary